Vijay Prajwal Rajasekar

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TECHNICAL SKILLS

CAD & Design Software: AutoCAD, SolidWorks & 3D-Printing, Onshape CAD, Microsoft word, Excel, PowerPoint, and Outlook, DraftSight

Manufacturing & Programming: PDM/ERP Systems, Time Studies & Downtime Analysis, Tool & Die and Jigs & Fixture Design, PLC-HMI Programming, Robot Controls: Fanu, Kuka, ABB, & Denso, Python, Power BI

Certifications: Certified SolidWorks Professional (**CSWP**) – 2022; Certified SolidWorks Associate (**CSWA**) – 2020

EDUCATION

University of British Columbia Expected April 2029

Bachelor of Applied Science – Mechanical Engineering - Aerospace Option

- Co-op Availability: 4 months beginning May 2027

Conestoga College Graduated

Mechanical Engineering Technology – Automated Manufacturing (Co-op), Cambridge, ON

- Relevant Coursework: Advanced SolidWorks (Sheet Metal, Weldment), PLC & HMI, Robot Controls Programming, CNC Programming (2D/3D), Additive Manufacturing, and Jigs & Fixture Design

TECHNICAL WORK EXPERIENCE

Filtervac International Inc. | Breslau, ON January 2023 - May 2025

Mechanical Engineering Technologist – Full-time

- Modeled 3D CAD components, engineering drawings, and exploded assembly view in SolidWorks to guide manufacturing and assembly on the shop floor
- Generated Bills of Material for the manufacturing of parts and machines, sending it to project manager for approval and purchasing
- Created models and drawings to support the manufacturing process, improving production efficiency and repeatability
- Prepared as-built drawings and costing reports for components, supporting accurate project documentation
- Inspected mechanical installations and conducted functional testing of machines, validating build quality against design specifications

Linamar Corporation (Skyjack Inc.) | Guelph, ON September 2021 - December 2021

Manufacturing Engineer Co-op II

- Timed welding-line operations in Excel and reviewed process and layout changes with supervisor
- Designed welding jigs in SolidWorks, keeping in mind of manufacturing cost and welder ease of use, built in-house for the welding station
- Documented all the tooling across assembly lines for a new facility (photos, part numbers, specs & quantities). Also installed a new tooling board for assembly lines
- Prepared Operation Availability (OA) reports, Lean suggestions, and paint-line work; used ERP (ManuX) to check materials and update machine tags

Linamar Corporation (Skyjack Inc.) | Guelph, ON January 2021 - April 2021

Manufacturing Engineer Co-op I

- Wrote assembly and welding work instructions from operator observations, with photos, approved for the shop floor procedures. This reduced the assembly and welding error
- Measured and modeled a physical assembly fixture to design SolidWorks drawings, and got approved for fabrication
- Made daily Operational Availability (OA) reports on Power BI software and ran downtimes studies on automated welding stations
- Audited shopfloor on weekly basis and creating a practical lean-suggestions to be implemented
- Automated a paint-line chemical calculator with Excel VBA
- Updated plant layout in DraftSight CAD software to keep it up to date

ENGINEERING DESIGN TEAMS

UBCO Aerospace Club | Kelowna, BC September 2025 - Present
Mech/Manf Sub - Team Lead

- 2026-2027 Project: Working on the AIAA Design/Build/Fly (DBF) Competition
- Contributing to the design and manufacturing of aircraft across the club's Fixed-Wing and UAV divisions
- Fixed-Wing Division: Designed structural components for the team's "Guardian" aircraft, including the fuselage and rear V-tail geometry (Onshape CAD)
- UAV Division: Led the manufacturing of carbon fiber composite components, designing mould tooling and developing the complete layup-to-cure fabrication procedure (SolidWorks, composite manufacturing)
- Contributed hands-on to the physical build and assembly of the Guardian aircraft, translating CAD designs into a flight-ready structure

TECHNICAL PROJECTS

Forged Carbon Fiber Motor Mount Project | UBCO Aerospace
(Mold Design & Layup Lead)

- Built and modeled drop-in replacement motor mounts in SolidWorks, matching existing bolt patterns and geometry to maintain full airframe compatibility
- Engineered and iterated a 4-stage multi-part manufacturing mold, optimizing wall draft angles, resin vent pathways, and parting-line clamp points
- Fabricated structural components using chopped-tow carbon fiber and resin hand-layup techniques, diagnosing and resolving defects across iterations
- Achieved a 57.9% mass reduction (114g to 48g per mount), removing 264g of structural dead weight across four airframe motors

Involute Gear Train | Personal Project, UBCO Makerspace

- Modeled a parametric, equation-driven involute gear tooth profile in SolidWorks
- Built a 4-gear compound train (12T-30T-14T-28T) achieving an overall 1:5 reduction ratio
- 3D printed all four gears using PLA on a Prusa i3 MK3S, adjusting tooth clearance to prevent gear tooths grinding and have free rotation
- Produced a functional, hand-driven gear train demonstrating compound gear ratios and speed reduction

PROFESSIONAL DEVELOPMENT

Technobytes Technology | IIT-M Research Park, Chennai, India
Drone Technology Workshop

- Completed a two-day, hands-on Drone Technology Workshop covering practical demonstrations and applied drone systems training
- Earned a certification of completion for the program

VOLUNTEER EXPERIENCE

Conestoga College | Kitchener, ON 2020
Orientation Volunteer

- Guided groups of 20 first-year students through campus tours as part of new student orientation
- Answered student questions regarding registration, timetables, and parking, supporting a smooth transition to campus life

AWARDS

SolidWorks Sprint – Best 1st Year Design | University of British Columbia Okanagan 2025

- Awarded jointly with a teammate; presented by Prof. Dr. Ray Taheri